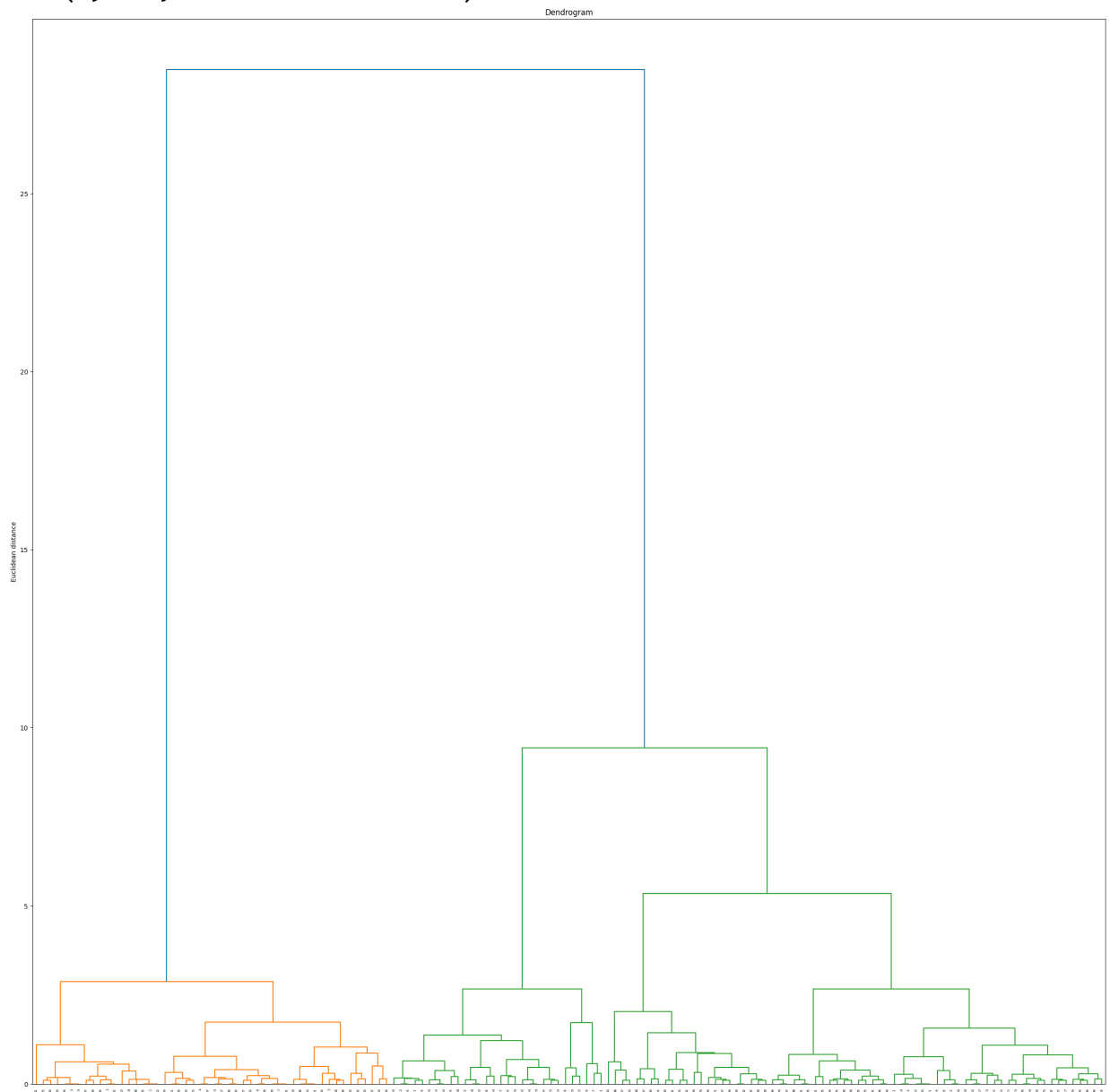


```
from sklearn.datasets import load_iris
from sklearn.cluster import AgglomerativeClustering
import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage
```

```
data = load_iris()
df = data.data
df = df[:,1:3]
Z = linkage(df, method = "ward")
plt.figure(figsize=(30,30))
dendro = dendrogram(Z)
plt.title("Dendrogram")
plt.ylabel("Euclidean distance")
```

```
Text(0, 0.5, 'Euclidean distance')
```



*
Q. What is Similarity based clustering

→ It is a technique in which data points are grouped into clusters based on how similar they are to each other. It uses measures such as Euclidean distance, cosine similarity or Jaccard distance index to determine closeness between objects. Points with high similarity are placed in same cluster, while dissimilar points fall into different cluster.

Q. Why do we need to perform significance testing in clustering.

→ It is important to ensure that the identified clusters represent genuine patterns in data rather than random groupings. Since clustering is unsupervised and lacks predefined labels, there is a risk of interpreting noise as structure. By performing significance testing, we can validate the quality and distinctiveness of clusters.

Q. Give example of confusion clustering to solve real-life problem.

→ It can be applied in healthcare for disease diagnosis. Patients' symptoms and medical test results may overlap across different diseases, creating confusion in classification. By using confusion clustering, patients of some patterns can be grouped together.

- a. difference between hard and soft clustering
- • Hard clustering: Each data point has exactly one cluster. No overlap, hence a point in a cluster, it cannot belong to another.
 - Soft clustering: Each data point can belong to multiple clusters with a certain probability or degree of membership.

- a. What are challenges associated with clustering
- There are multiple challenges:
 1. Choosing no. of clusters in advance
 2. Selecting an appropriate similarity measure
 3. Handling high-dimensional data
 4. Dealing with noise and outliers
 5. Scalability to large datasets.

Q.